

Ayan Shaikh

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Education

Wilfrid Laurier University

Sept 2023 – Apr 2027

Honors BSc, Computer Science with an Economics Minor

Relevant Coursework: Data Structures and Algorithms, Databases I, Computer Networks, Software Engineering, Digital Electronics, Introduction to Microprocessors, Stats & Probabilities, Linear Algebra, Discrete Structures

Skills

Languages: Python, Java, C, C++, C#, HTML, CSS, JavaScript, Swift, SQL

Technologies: NumPy, Pandas, MongoDB, MySQL, Linux, Git, Microsoft Office

Frameworks/Libraries: React.js, Node.js, Three.js, Express.js

Professional Experience

WELL Health Diagnostic Centre

Milton, ON

Patient Care Representative

Sept 2024 – Current

- Supported a clinic-wide systems migration during a corporate merger by **configuring Microsoft Office items** and transferring data, successfully onboarding **15+** staff while ensuring 100% uptime in patient communication
- Streamlined patient intake and scheduling workflows, reducing average wait times by **17%** and increasing satisfaction scores by **20%** over three months through collaboration with medical and administrative teams.
- Delivered high-quality service to **200+** patients weekly through multiple communication channels, managing appointments and insurance verification to ensure seamless access to diagnostic and immigration x-ray services

Projects

Client-Server Bulletin Board System | Java, TCP/IP Sockets, Java Swing, Network Programming

Feb 2026

- Implemented a threaded **Java** server using **ServerSocket** and per-client handlers to manage concurrent clients interacting with a shared bulletin board.
- Designed a custom application-layer protocol over **TCP**, including structured responses, command parsing, and robust error codes.
- Created a **real-time GUI** client in **Java Swing** featuring automatic polling, selective refresh pausing, visual note rendering, and multi-client synchronization.

3D Render Engine | Java, Java Swing, Object-Oriented Programming

Jan 2026

- CPU-based 3D rendering engine in **Java** from scratch, implementing the full graphics pipeline including vertex transformations, triangle rasterization and z-buffering, without relying on OpenGL or other third-party libraries
- Developed custom linear algebra and lighting system, including 3x3 matrix transformations, face normal computation and gamma-corrected **Lambert shading** with a real-time camera control through an interactive **Swing UI**

SneakerHub | JavaScript, Tailwind CSS, HTML

Nov 2025

- Built a responsive e-commerce platform with dynamic product pages, search filtering, and interactive UI components using **HTML**, **Tailwind CSS**, and **JavaScript**.
- Implemented client-side state management with **localStorage** to handle user authentication, cart persistence, selective checkout, order placement, order status updates (processing, shipped, cancelled), and order history without a backend.
- Applied modular **JavaScript patterns** and reusable UI components to maintain consistency across navigation, authentication handling, and shared application state.

MMA Database | MySQL, React.js, Express.js, Node.js

Apr 2025

- Designed and implemented a fully normalized MMA relational database (BCNF)** modeling fighters, events, fights, contracts, training camps, and sponsorships, enforcing data integrity through primary/foreign keys and functional dependency analysis.
- Developed and optimized **10 advanced SQL queries** and **3 views** using multi-table joins, subqueries, aggregations, and window functions to analyze fighter performance, rankings, sponsorship totals, and organizational statistics.
- Built a **full-stack web application** to enable dynamic querying and visualization of MMA data, demonstrating end-to-end database integration from schema design to frontend consumption.